Praneeth Tarlapalli

AI/ML Engineer

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SUMMARY

Applied AI & Machine Learning Engineer with 5 years of experience architecting, scaling, and deploying enterprise-grade artificial intelligence systems, multi-agent frameworks, and high-performance data pipelines. Expert in designing stateful multi-agent orchestrations, enterprise RAG pipelines, and fine-tuning domain-specific frontier models using tools like LangGraph, LlamaIndex, and Azure AI Foundry. Proven track record across the full ML lifecycle—from deep learning and NLP development to high-throughput API deployment via FastAPI, Docker, and Kubernetes across Azure, AWS, and GCP. Combines robust data engineering foundations in PySpark, SQL, and Apache Airflow with disciplined production monitoring, specializing in model explainability, risk evaluation, and automated drift detection to maintain model integrity in enterprise environments.

SKILLS

Programming & Querying: Python, SQL, R
Generative AI & LLM Frameworks: Hugging Face Transformers (BERT, GPT, T5), LangChain, LlamaIndex, LangGraph, Prompt Engineering, Retrieval-Augmented Generation (RAG), AutoGen, Haystack, Model Fine-Tuning (LoRA, QLoRA), Claude Code, OpenAI Codex, Cursor, Context Engineering, Agentic Workflows, Multi-Agent Orchestration
Data Analytics & Statistics: Pandas, NumPy, SciPy, Statistical Analysis, A/B Testing, Time Series Analysis, Predictive Analytics
Cloud Platforms & Deployment: AWS (EC2, S3, SageMaker, Lambda), Azure (Azure ML, Azure Data Factory, Azure Blob Storage), GCP (BigQuery, Vertex AI, Cloud Storage), Docker, Kubernetes, Streamlit, Cloud ML Pipelines, vLLM, SGLang
Machine Learning & AI: Scikit-learn, TensorFlow, Keras, Deep Learning (CNN, RNN, LSTM, GRU), Natural Language Processing (NLP), Feature Engineering, Hyperparameter Tuning, Model Explainability (SHAP, LIME)
Data Engineering & Real-Time Processing: Apache Kafka, Apache Spark, PySpark, Apache Airflow, MLflow
Data Visualization & Business Intelligence: Tableau, Power BI, Matplotlib, Seaborn, Plotly
Monitoring & Production Readiness: Model Performance Monitoring, Data Drift Detection, Concept Drift Detection
Computer Vision: OpenCV, Object Detection (YOLO, Ultralytics, Faster R-CNN), Image Segmentation (U-Net, Mask R-CNN, SAM), Image Classification (ResNet, EfficientNet, Vision Transformers (ViT), Generative Vision
APIs & Backend Development: FastAPI, Flask, RESTful APIs
Databases & Vector Stores: MySQL, PostgreSQL, Oracle, MongoDB, Data Warehousing Concepts, FAISS, Pinecone, ChromaDB, Milvus, Weaviate
Version Control & CI/CD: Git, GitHub, GitHub Actions, GitLab, GitLab CI/CD, Bitbucket

PROFESSIONAL EXPERIENCE

Applied AI Engineer — Feb 2026 – Present | Austin, TX
KPMG

- Built stateful multi-agent orchestration workflows using Azure AI Foundry and LangGraph within the internal *KPMG Workbench* platform, configuring autonomous reasoning loops to manage parallel financial forecasting and simulation pipelines.
- Deployed high-throughput backend APIs using FastAPI and Azure Kubernetes Service (AKS) to ingest real-time telemetry streams and evaluate transaction risk profiles natively inside the *KPMG Clara* auditing ecosystem.
- Architected enterprise RAG pipelines via Microsoft Fabric, Azure AI Search, and ChromaDB, integrating custom SHAP-based evaluation steps and alignment guardrails that drove a 33% reduction in pre-implementation audit anomalies.
- Fine-tuned domain-specific frontier models via Azure OpenAI Service and Azure Machine Learning to parse legal taxonomy, optimizing context windows within the *Digital Gateway* platform to achieve a 22% gain in precise legal retrieval.
- Built predictive quality control models using Scikit-learn and PySpark, processing telemetry from Azure IoT Hub edge nodes to sync live asset failure predictions directly into factory workflow digital twins.

ML Engineer — Jan 2025 – Nov 2025 | Sugar land, TX
Hillsoft Business Solutions

- Designed and deployed end-to-end predictive analytics models using Scikit-learn, Pandas, and NumPy, conducting detailed feature engineering and statistical analysis to extract insights from raw client datasets.
- Built and automated scalable machine learning workflows using Apache Airflow and PySpark, streamlining enterprise-level client data transformation, validation, and feature storage pipelines.
- Developed high-throughput backend APIs using FastAPI and Flask to serve live model inferences, containerizing the entire application stack using Docker for standardized environment deployment.
- Executed rigorous hyperparameter tuning, cross-validation, and A/B testing cycles across models deployed on AWS (EC2, S3), reducing model inference latency by 18%.

- Created interactive diagnostic and performance dashboards using Power BI, Matplotlib, and Seaborn to translate complex predictive model metrics into actionable business intelligence for stakeholder reporting.
- Integrated automated model performance monitoring systems utilizing MLflow, implementing dedicated data and concept drift detection mechanisms to maintain baseline prediction degradation below 4%.

Cognizant

Data Scientist I — Jul 2022 – Oct 2023 | Hyderabad, India

- Built text classification and entity extraction models by leveraging Hugging Face Transformers (BERT, T5) and Scikit-learn, which drove a 15% efficiency gain in automating enterprise document processing.
- Wrote optimized SQL scripts and PySpark data pipelines to aggregate and clean millions of rows of unstructured text, establishing reliable feature engineering baselines for the analytics team.
- Trained deep learning architectures, focusing on LSTM and GRU networks for time-series forecasting, and packaged the models into lightweight Flask RESTful APIs for application integration.
- Configured training jobs within AWS SageMaker to run hyperparameter tuning and model evaluation, optimizing model architecture to trim inference latency by 20%.
- Designed interactive performance tracking dashboards using Power BI, Matplotlib, and Seaborn to visualize statistical variations, validation metrics, and model predictive drift for business teams.
- Maintained production model integrity by configuring MLflow tracking servers, building automated pipelines to monitor incoming data streams for data and concept drift.

Data Engineer — Jan 2020 – Jun 2022 | Hyderabad, India

- Built robust, cloud-native ETL/ELT pipelines using GCP Dataflow and Cloud Composer to ingest and process massive relational and streaming datasets into central repositories.
- Optimized data warehouse structures inside Google Cloud BigQuery, configuring partitioning, clustering, and materialized views to decrease complex query latency by 25%.
- Wrote scalable Python and PySpark batch data transformation scripts to aggregate fragmented corporate data, enforcing data validation logic and reducing ingestion errors.
- Architected automated CI/CD deployment pipelines using Cloud Build and GitHub Actions, establishing consistent and repeatable delivery setups across staging and production data environments.
- Developed logging, automated alerting, and pipeline performance monitoring frameworks, maintaining overall cloud data infrastructure uptime above 99.9%.

EDUCATION

Master's in Computer Science — Jan 2024 – Dec 2025 | Magnolia, AR
Southern Arkansas University

Bachelor's in Information Technology (IT) — Aug 2017 – May 2021 | Bangalore, India
NEW horizon college